

(Vapour + liquid) equilibria and excess molar enthalpies for mixtures with strong complex formation. Trichloromethane or 1-bromo-1-chloro-2,2,2-trifluoroethane (halothane) with tetrahydropyran or piperidine

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Isothermal (vapour + liquid) equilibria were measured for (trichloromethane + tetrahydropyran or piperidine) at $T = 333.15$ K and {1-bromo-1-chloro-2,2,2-trifluoroethane (halothane) + tetrahydropyran or piperidine} at $T = 323.15$ K with a circulation still. The results were verified by effective statistical procedures and used to calculate activity coefficients and excess molar Gibbs free energies G_m^E . Excess molar enthalpies H_m^E for these mixtures were determined at $T = 298.15$ K by means of an isothermal CSC microcalorimeter equipped with recently reconstructed flow mixing cells. Reliable performance of the calorimetric setup was proved by the good agreement of H_m^E for (hexane + cyclohexane), (2-propanone + water), and (methanol + water), with the best literature results. The trichloromethane- or halothane-containing mixtures exhibit strong negative deviations from Raoult's law and are highly exothermic, thus indicating that complex formation via hydrogen bonding is a governing nonideality effect. A close similarity in the behaviour of corresponding mixtures with trichloromethane and halothane is observed, but for halothane-containing mixtures, G_m^E and H_m^E are consistently more negative, confirming that halothane is a more powerful proton donor than chloroform.

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1. Introduction

Due to the highly acidic hydrogen atom in the molecules of trichloromethane and halothane, strong complex formation via hydrogen bonding occurs when these halocarbons

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TABLE 1. Densities and vapour pressures of pure compounds

Compound	$\rho/(\text{g} \cdot \text{cm}^{-3}), T = 298.15 \text{ K}$			T/K	$p^*(\text{kPa})$		
	expt	lit.	ref.		expt	lit.	ref.
CF_3CHBrCl	1.8560	1.85631	13	323.15	100.41	100.41	14
CHCl_3	1.4795	1.4797	14	333.15	97.37	97.41	15
$c\text{-(CH}_2)_5\text{O}$	0.8792	0.87916	16	323.15	27.32	27.51	17
		0.87911	18	333.15	39.81	40.09	17
$c\text{-(CH}_2)_5\text{NH}$	0.8569	0.8568	19	323.15	13.43	13.52	20
				333.15	20.58	20.59	20

are mixed with efficient proton-acceptors. We have studied this interesting class of mixtures within a long-term project on the basis of a variety of thermodynamic properties {(vapour + liquid) equilibria,^(1,2) limiting activity coefficients,⁽³⁾ excess molar enthalpies,^(4,5) excess molar heat capacities, and excess molar volumes^(5,6)} and the ^1H n.m.r. spectroscopic technique.^(7,8) Analysis of the accumulated experimental information through association models enabled us to give quantitative answers to questions concerning the degree of complex formation and the stability of complexes.^(2,9,10) In this work, the list of proton-acceptors we have dealt with so far (tetrahydrofuran, propyl ether, isopropyl ether, *tert*-butyl methyl ether, 1,4-dioxane, 2,5-dioxahexane, 2-propanone, methyl ethanoate, and methanol) is further extended by tetrahydropyran and piperidine, for mixtures of which with trichloromethane and halothane we report here careful measurements of (vapour + liquid) equilibria and excess molar enthalpies.

2. Experimental

Trichloromethane of analytical-reagent grade (Lachema, Brno, Czech Republic) stabilised with 1 mass per cent of ethanol was first washed with water to remove the stabiliser and then dried with calcium chloride. The substance was further refluxed with a small portion of 2,4-dinitrophenylhydrazine acidified with hydrogen chloride and then fractionally distilled under nitrogen twice, on 1 m and 1.5 m columns packed with ceramic saddles. Tetrahydropyran, puriss (Fluka), which was first distilled on a 0.5 m packed column, halothane of medical grade without a stabiliser (Léčiva, Praha, Czech Republic), and piperidine, puriss (Fluka) were all dried with 0.4 nm molecular sieves and stored in the dark in sealed ampoules, while trichloromethane was purified and used immediately for measurements because of its rapid decomposition. Methanol, 2-propanone, hexane, and cyclohexane were all analytical-reagent grade chemicals from Penta (Chrudim, Czech Republic) and were used as purchased. The purity of each organic chemical used was checked by g.c. and found to be mass fraction >0.999 . Water was distilled and deionized in a Milli-Q RG Water Purification System (Millipore, U.S.A.). Prior to the calorimetric measurements, all liquids were degassed by heating and subsequent low-pressure filtering through fine-porosity frits.

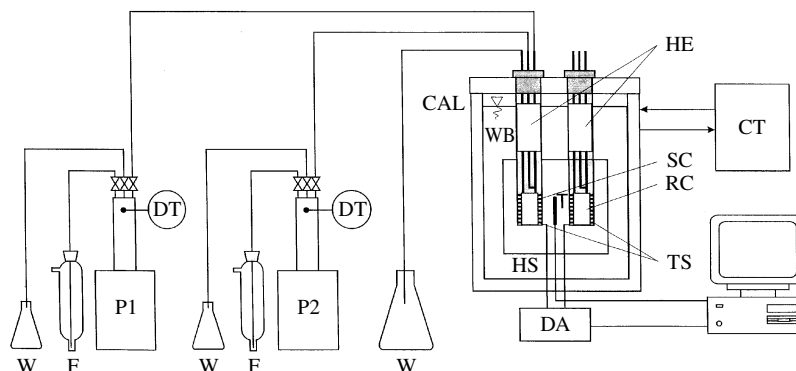


FIGURE 1. Schematic diagram of the flow mixing microcalorimeter setup. CAL, microcalorimeter; SC, sample cell; RC, reference cell; HE, heat exchanger; HS, heat sink; TS, thermoelectric sensors; T, thermometer; DA, differential amplifier; CT, cooling thermostat; WB, water bath; P1, P2, HPLC syringe pumps; DT, digital thermometer; F, feed bottle; W, waste bottle.

The (vapour + liquid) equilibria were measured in a modified Dvořák-Boublík circulation still described in detail previously.⁽¹¹⁾ The pressure was controlled by a two-stage on-off manostat connected to a Texas Instruments Model 145 precision pressure gauge. The short-term reproducibility of the measurement and pressure setting was about ± 1 Pa, and the estimated absolute accuracy was ± 7 Pa. The temperature was measured with a Hewlett-Packard 2801A quartz-crystal thermometer with a 2850A probe and recorded on a digital printer. The thermometer was calibrated *in situ* against several ebulliometric standards. The reproducibility of temperature measurement was within a few mK, and the absolute accuracy is estimated to be ± 20 mK. The compositions of the equilibrium mixtures were calculated from densities measured at $T = 298.15$ K, on the basis of measured dependences of the excess molar volume on composition. The densities of the equilibrium phases were determined by means of an Anton Paar vibrating-tube density meter, Model DMA 5000. The molar volume V_m^E measurements were carried out by us previously on a Sodev vibrating-tube density meter and are reported elsewhere.⁽¹²⁾ Vapour pressures and densities of pure components used in (vapour + liquid) determinations, as measured with the equipment described above, are compared with literature values in table 1. Agreement is generally very good except for the vapour pressure of tetrahydropyran where, however, no measurements other than those of reference 17 were found in the literature to enable us to resolve the discrepancy.

Excess molar enthalpies were measured with an isothermal microcalorimeter Model 4400 manufactured by Calorimetry Science Corp. (CSC), Provo, U.S.A. This instrument of differential heat conduction design incorporates two test wells (sample and reference) in a large aluminium heat sink that is immersed in an ultra-stable thermostating bath. The microcalorimeter is equipped with flow mixing cells (Model 4442) that have recently been reconstructed by CSC to improve their performance. Contrary to the original construction which used an inefficient mixing tee, mixing is achieved in the new design in a concentric

TABLE 2. Experimental (vapour + liquid) equilibria; liquid-phase mole fraction x , vapour-phase mole fraction y , vapour pressure p , activity coefficients γ_1 and γ_2 , and excess molar Gibbs free energy G_m^E

x	y	p/kPa	γ_1	γ_2	$G_m^E/(\text{J} \cdot \text{mol}^{-1})$
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{O}; T = 333.15 \text{ K}$					
0	0	39.805	0	1	0
0.0298	0.0263	39.671	0.3620	1.0003	-83.1
0.0736	0.0698	39.487	0.3878	0.9962	-203.0
0.1363	0.1423	39.475	0.4276	0.9847	-357.6
0.2150	0.2499	39.971	0.4832	0.9586	-525.2
0.3149	0.4071	41.851	0.5641	0.9066	-685.5
0.4242	0.5845	46.030	0.6619	0.8269	-788.0
0.5308	0.7365	52.532	0.7603	0.7285	-814.6
0.6244	0.8397	60.408	0.8457	0.6307	-769.5
0.6825	0.8867	65.713	0.8873	0.5701	-720.2
0.6903	0.8922	66.669	0.8953	0.5637	-703.3
0.7101	0.9054	68.870	0.9117	0.5445	-669.7
0.7631	0.9344	74.069	0.9400	0.4942	-593.3
0.8453	0.9666	82.515	0.9751	0.4254	-425.2
0.9458	0.9910	92.343	0.9965	0.3625	-161.5
0.9767	0.9963	95.199	0.9991	0.3563	-69.0
1	1	97.366	1		0
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{NH}; T = 333.15 \text{ K}$					
0	0	20.584		1	0
0.0680	0.0925	21.066	0.2990	0.9961	-237.4
0.1242	0.1822	21.733	0.3332	0.9847	-415.5
0.1896	0.3006	22.947	0.3809	0.9594	-599.9
0.3223	0.5607	27.854	0.5083	0.8682	-869.4
0.3872	0.6723	31.660	0.5766	0.8092	-949.8
0.4545	0.7687	36.910	0.6542	0.7419	-985.2
0.5160	0.8371	42.607	0.7233	0.6739	-992.1
0.5754	0.8866	48.847	0.7861	0.6072	-970.3
0.6412	0.9249	56.325	0.8465	0.5427	-903.5
0.6831	0.9441	61.755	0.8875	0.4975	-838.4
0.7519	0.9659	70.038	0.9330	0.4344	-717.4
0.8673	0.9876	83.848	0.9853	0.3468	-424.9
0.9424	0.9959	91.727	0.9976	0.2858	-206.1
1	1	97.366	1		0

TABLE 2—*continued*

x	y	p/kPa	γ_1	γ_2	$G_{\text{m}}^{\text{E}}/(\text{J} \cdot \text{mol}^{-1})$
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}; T = 323.15 \text{ K}$					
0	0	27.322		1	0
0.0257	0.0268	27.332	0.2815	0.9992	−89.5
0.0554	0.0606	27.388	0.2967	0.9967	−189.1
0.0988	0.1157	27.578	0.3211	0.9898	−326.4
0.1462	0.1839	27.976	0.3515	0.9770	−464.3
0.2045	0.2778	28.848	0.3935	0.9545	−612.0
0.2644	0.3843	30.271	0.4440	0.9193	−742.9
0.3445	0.5320	33.311	0.5218	0.8548	−878.3
0.5171	0.7941	45.599	0.7125	0.6733	−984.1
0.5896	0.8635	53.116	0.7900	0.5989	−938.5
0.6601	0.9140	61.371	0.8605	0.5145	−873.2
0.7410	0.9505	71.273	0.9222	0.4397	−732.9
0.8115	0.9721	79.767	0.9604	0.3728	−587.8
0.8674	0.9837	86.364	0.9816	0.3297	−438.7
0.9317	0.9933	93.771	0.9986	0.2806	−236.7
0.9640	0.9967	96.849	0.9989	0.2687	−130.0
1	1	100.411	1		0
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{NH}; T = 323.15 \text{ K}$					
0	0	13.429		1	0
0.0353	0.0592	13.720	0.2332	0.9961	−148.3
0.0792	0.1415	14.280	0.2593	0.9903	−311.4
0.1304	0.2475	15.159	0.2932	0.9740	−491.4
0.2146	0.4337	17.545	0.3628	0.9339	−728.9
0.3880	0.7602	27.688	0.5565	0.7778	−1024.2
0.4289	0.8131	31.479	0.6117	0.7304	−1048.3
0.5288	0.9016	42.714	0.7436	0.6124	−1041.6
0.5661	0.9230	47.321	0.7864	0.5690	−1022.8
0.6006	0.9394	51.912	0.8260	0.5269	−995.9
0.6213	0.9468	54.612	0.8457	0.5095	−965.9
0.7049	0.9699	65.777	0.9152	0.4322	−832.9
0.7585	0.9791	72.691	0.9460	0.3978	−711.2
0.7982	0.9846	77.622	0.9632	0.3697	−619.8
0.8365	0.9890	82.424	0.9782	0.3417	−521.2
0.9063	0.9947	90.511	0.9936	0.3088	−311.4
0.9641	0.9984	96.768	0.9996	0.2559	−132.5
1	1	100.411	1		0

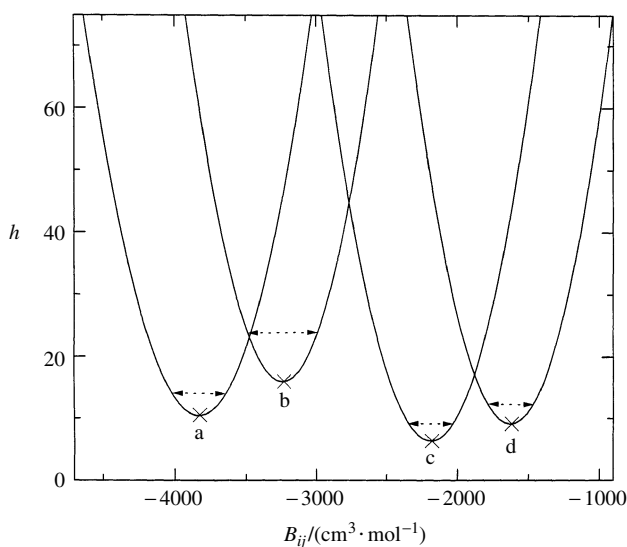


FIGURE 2. Dependence of the consistency criterion h on the second cross virial coefficient B_{ij} for a, $\{x\text{CF}_3\text{CHBrCl} + (1-x) \text{c}-(\text{CH}_2)_5\text{NH}\}$; b, $\{x\text{CF}_3\text{CHBrCl} + (1-x) \text{c}-(\text{CH}_2)_5\text{O}\}$; c, $\{x\text{CHCl}_3 + (1-x) \text{c}-(\text{CH}_2)_5\text{NH}\}$; d, $\{x\text{CHCl}_3 + (1-x) \text{c}-(\text{CH}_2)_5\text{O}\}$; x, maximum likelihood estimate $\hat{B}_{ij}$, $\leftarrow - - - \rightarrow$, 95 per cent confidence interval for $\hat{B}_{ij}$.

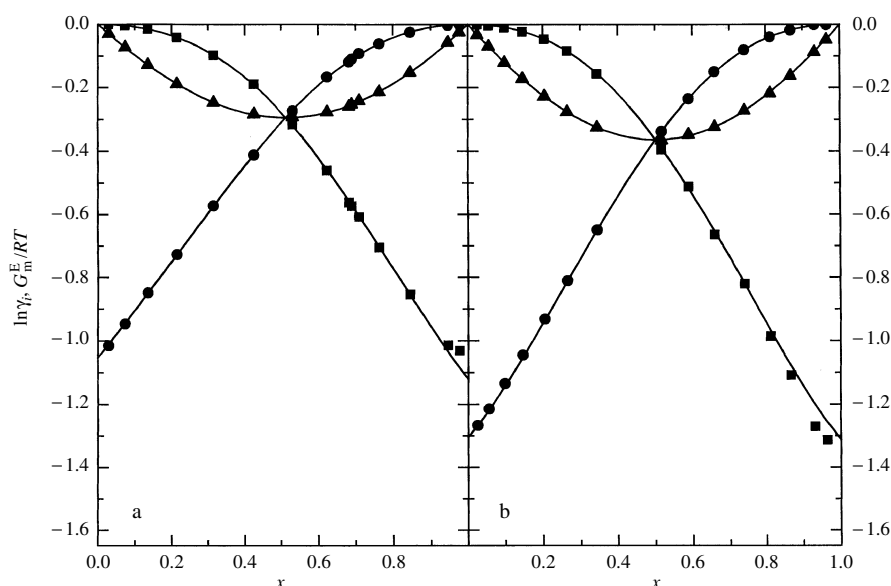
tube arrangement in which the inner tube and the outer tube provide the inlets for two liquids to be mixed.

A schematic diagram of the flow mixing microcalorimeter assembly is given in figure 1. Two high-pressure liquid chromatography syringe pumps. Model HPP 5001 by Laboratorní Pístroje (Praha, Czech Republic), were employed to inject the components into the calorimeter. The components are continuously delivered into the sample cell at a constant total flow rate chosen from the range $(1 \text{ to } 8) \text{ cm}^3 \cdot \text{s}^{-1}$, while the ratio of individual flow rates is varied to carry out measurements for different compositions. The flow rates can be set with a resolution of about $0.15 \text{ cm}^3 \cdot \text{s}^{-1}$, and the temperature of the fluid in the pump cylinder is measured with a digital thermometer. Calibration of the pump flow rates was done by flowing water through the system while timing and then weighing the delivered amount. Replicates showed flow-rate reproducibility to be within 0.3 per cent. The reference cell used to compensate for electronic noise and any heat flux due to temperature fluctuations in the heat sink is left empty. The differential signal from the calorimeter test wells thus corresponds to the rate of heat production from the sample cell itself. The signal is calibrated by means of the Joule effect produced by a built-in calibration heater on the pure liquid (before and after each set of experiments). Data acquisition and calibration is controlled by a PC.

In preliminary experiments, the performance of the calorimeter was examined on the test system (hexane + cyclohexane). The measurements carried out at three flow ratios as a function of the total flow rate F through the calorimeter showed that the measured

TABLE 3. Second virial coefficients B_{ii} , B_{jj} and B_{ij} and solvation parameters η_{ij} for the Hayden–O’Connell method

	T/K	$\frac{B_{ii}}{\text{cm}^3 \cdot \text{mol}^{-1}}$	$\frac{B_{jj}}{\text{cm}^3 \cdot \text{mol}^{-1}}$	$\frac{B_{ij}}{\text{cm}^3 \cdot \text{mol}^{-1}}$	η_{ij}
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{O}$	333.15	−894	−1255	−1610	1.2
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{NH}$	333.15	−894	−1683	−2180	1.8
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}$	323.15	−1088	−1366	−3230	2.05
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{NH}$	323.15	−1088	−1839	−3790	2.25

FIGURE 3. Values of $\ln \gamma_1$, $\ln \gamma_2$ and G_m^E/RT plotted against mole fraction x for a, $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{O}\}$ at $T = 333.15$ K and b, $\{x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}\}$ at $T = 323.15$ K.

values of H_m^E dropped slightly with increasing F . In the examined range of F from (0 to $8 \text{ cm}^3 \cdot \text{s}^{-1}$), the relative drop of the measured values of H_m^E was found to be proportional to F with a uniform slope $(\partial H_m^E / \partial F) / H_m^E = 6 \cdot 10^{-3} \text{ s} \cdot \text{cm}^{-3}$, the extrapolated value of H_m^E at $F = 0$ being, for each of the three flow ratios, in excellent agreement with reliable literature values.^(21–23) The correction for non-zero flow rates thus established was applied to all subsequent measurements.

3. Results and discussion

The (vapour + liquid) equilibria were determined isothermally at $T = 333.15$ K for trichloromethane mixtures and at $T = 323.15$ K for halothane mixtures. The results are

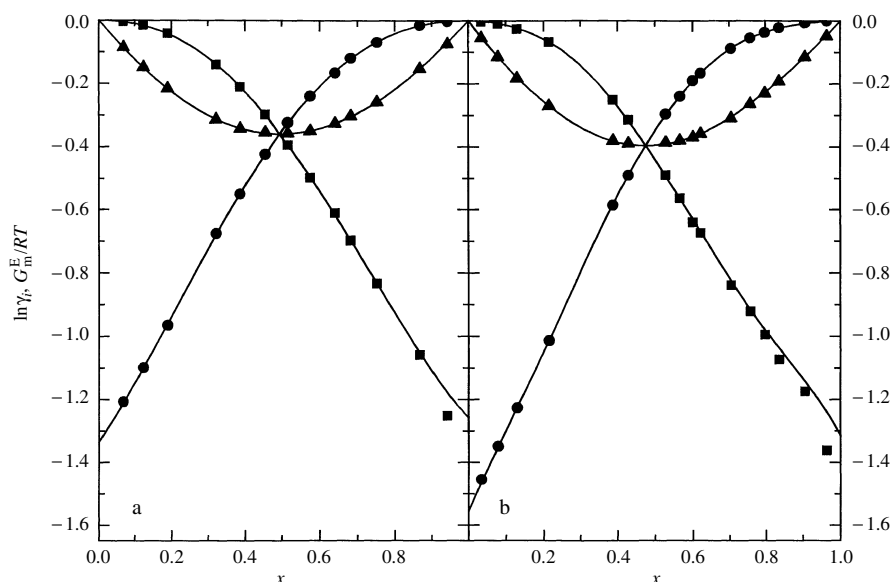


FIGURE 4. Values of $\ln \gamma_1$, $\ln \gamma_2$ and G_m^E/RT plotted against mole fraction x for a, $\{x\text{CHCl}_3 + (1-x)\text{c}-(\text{CH}_2)_5\text{NH}\}$ at $T = 333.15$ K and b, $\{x\text{CF}_3\text{CHBrCl} + (1-x)\text{c}-(\text{CH}_2)_5\text{NH}\}$ at $T = 323.15$ K.

listed in table 2, along with the activity coefficients γ_i and excess molar Gibbs energies G_m^E calculated from the relations:

$$\ln \gamma_i = \ln(y_i p / x_i p_i^*) + (B_{ii} - V_{m,i}^l)(p - p_i^*)/RT + p\delta_{ij}(1 - y_i)^2/RT, \quad (1)$$

$$\delta_{ij} = 2B_{ij} - B_{ii} - B_{jj}, \quad (2)$$

$$G_m^E = RT(x_1 \ln \gamma_1 + x_2 \ln \gamma_2), \quad (3)$$

where x_i and y_i denote the mole fractions of the i th component in the liquid and vapour phases, respectively, p is the total vapour pressure, p_i^* is the vapour pressure of the pure component i , and $V_{m,i}^l$ is the molar volume of pure liquid i calculated from a modified Rackett equation.⁽²⁴⁾ The second virial coefficients B_{ii} , B_{jj} , and B_{ij} were estimated from the correlation of Hayden and O'Connell.⁽²⁵⁾ Pure-component parameters for the method were taken for halothane from our previous work⁽²⁶⁾ and for trichloromethane from Prausnitz *et al.*⁽²⁷⁾ For tetrahydropyran and piperidine, they were taken from CDATA,⁽¹⁴⁾ except for the association parameter η_{ii} that was approximated by the value for the closely related compounds, tetrahydrofuran and diethyl amine, respectively, and taken from Prausnitz *et al.*⁽²⁷⁾ The solvation parameters η_{ij} necessary for the estimation of B_{ij} by the Hayden and O'Connell method were not available, and hence we estimated the second cross-virial coefficients directly from measured (p, T, x, y) values by our model-free method.^(28,29) This method, based on the optimization of the thermodynamic consistency criterion h defined in reference 28, provides the best estimate of B_{12} in the

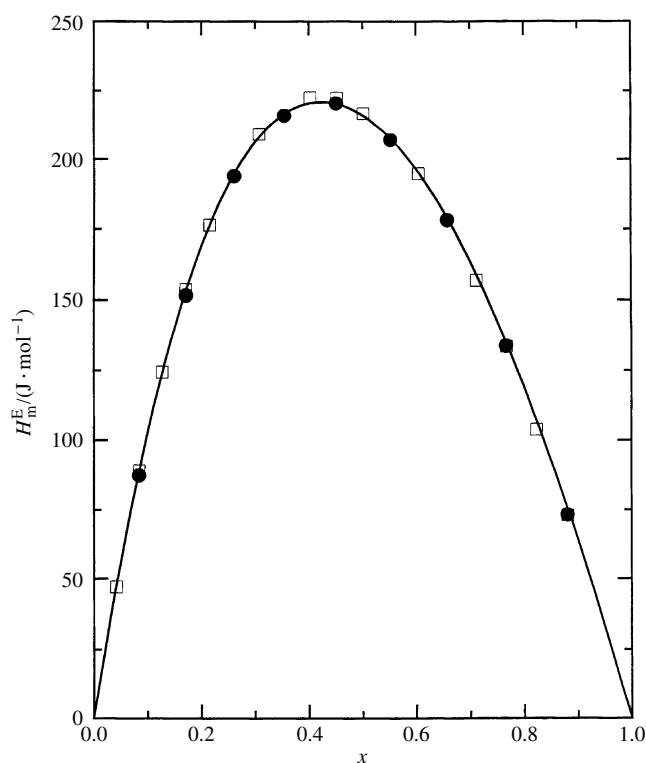


FIGURE 5. Excess molar enthalpies H_m^E at 298.15 K plotted against mole fraction x for $\{xC_6H_{14} + (1-x)c-C_6H_{12}\}$: ●, this work, total flow rate $3.3 \mu l \cdot s^{-1}$; □, this work, total flow rate $6.6 \mu l \cdot s^{-1}$; —, Marsh and Stokes.⁽²¹⁾

sense of maximum likelihood. The dependence $h(B_{12})$ for the four investigated mixtures is shown in figure 2, where the maximum likelihood estimates $\hat{B}_{12}$ and their 95 per cent confidence intervals are also indicated. The corresponding values η_{ij} and all the second virial coefficients used to reduce the (vapour + liquid) equilibrium data are listed in table 3.

Experimental uncertainties s in the determination of (vapour + liquid) equilibrium as checked by our test for consistency⁽²⁸⁾ and verification procedure⁽³⁰⁾ are $s(x) = s(y) = 0.0004$, $s(p) = 20$ Pa, and $s(T) = 20$ mK for all the studied mixtures. The (vapour + liquid) equilibrium measurements were found to be thermodynamically consistent (i.e., satisfying the Gibbs–Duhem equation) within the stated inaccuracies at a significance level of 0.05.

The measured (vapour + liquid) equilibria were correlated by the equation:

$$G_m^E/RT = x(1-x) \sum_{i=1}^q a_i (2x-1)^{(i-1)}. \quad (4)$$

The parameters a_i evaluated by optimization according to the principle of maximum likelihood are given in table 4 together with the standard deviation of fit. The number

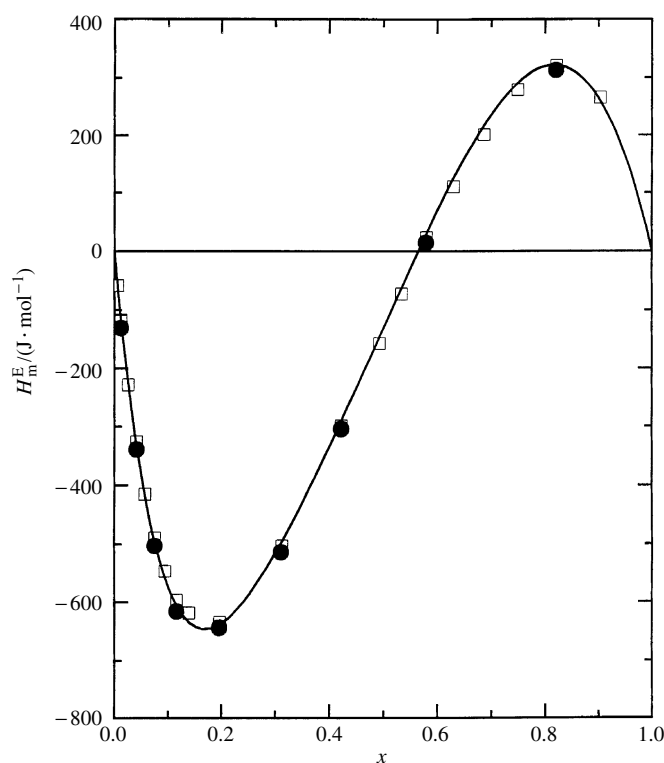


FIGURE 6. Excess molar enthalpies H_m^E at 298.15 K plotted against mole fraction x for $\{x(\text{CH}_3)_2\text{CO} + (1-x)\text{H}_2\text{O}\}$: ●, this work, total flow rate $3.3 \mu\text{l} \cdot \text{s}^{-1}$; □, this work, total flow rate $6.6 \mu\text{l} \cdot \text{s}^{-1}$; —, French.⁽³¹⁾

q of parameters in equation (4) was determined by the F-test, other statistical behaviour of residuals being also considered. The experimental values of G_m^E/RT , $\ln \gamma_i$ and their fits are shown in figures 3 and 4.

Excess molar enthalpies at $T = 298.15$ K were measured first for three test mixtures to verify the performance of the calorimeter. The mixtures (hexane + cyclohexane), (2-propanone + water), and (methanol + water) that were selected for the purpose were of different character and varied in magnitude and sign of their enthalpy-of-mixing effects. The results of our measurements are listed in table 5 and compared with the best measurements from the literature in figures 5 to 7. The agreement is excellent: for (hexane + cyclohexane) within 0.5 per cent and for the two other mixtures within 1.5 per cent of the maximum enthalpy-of-mixing effect. Note that for (hexane + cyclohexane) and (2-propanone + water) the present measurements were carried out at two different levels of the total flow rate (3.3 and 6.6) $\text{cm}^3 \cdot \text{s}^{-1}$, yet with coinciding results.

The experimental values of H_m^E at $T = 298.15$ K for the mixtures of our main interest, i.e. $\{(\text{trichloromethane or halothane}) + (\text{tetrahydropyran or piperidine})\}$, are given also in table 5 and are plotted in figures 8 and 9. Since for $\{x\text{CHCl}_3 + (1-x)\text{c}-(\text{CH}_2)_5\text{O}\}$

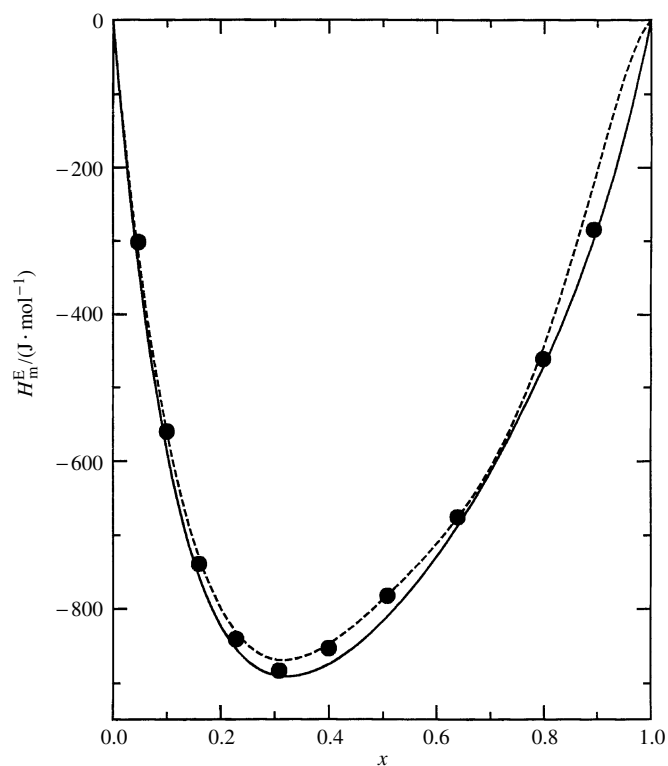


FIGURE 7. Excess molar enthalpies H_m^E at 298.15 K plotted against mole fraction x for $\{x\text{CH}_3\text{OH} + (1-x)\text{H}_2\text{O}\}$: ●, this work, total flow rate $3.3 \mu\text{l} \cdot \text{s}^{-1}$; —, Fenby and Chand⁽³²⁾; ---, Tomasziewicz *et al.*⁽³³⁾

TABLE 4. Parameters a_i and standard deviations s of fit for the representation of (vapour + liquid) equilibria by equation (4)

	T/K	a_1	a_2	a_3	a_4	a_5	s^a
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{O}$	333.15	-1.1792	-0.0612	0.0924	0.0271		1.14
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{NH}$	333.15	-1.4397	0.0382	0.1436			1.08
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}$	323.15	-1.4584	-0.0046	0.1502			1.21
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{NH}$	323.15	-1.5803	0.1839	0.2430	-0.0645	-0.0974	1.07

$$^a s = \{S/(2n - q)\}^{1/2}; S = \sum_{j=1}^n [\{\Delta x_j/s(x)\}^2 + \{\Delta y_j/s(y)\}^2 + \{\Delta p_j/s(p)\}^2 + \{\Delta T_j/s(T)\}^2].$$

TABLE 5. Experimental values of the excess molar enthalpy H_m^E at 298.15 K

x	$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}}$
$x\text{C}_6\text{H}_{14} + (1-x)\text{c-C}_6\text{H}_{12}$					
0.0416	47.0	0.3074	209.0	0.6578	178.1
0.0840	88.9	0.3548	215.9	0.7119	156.9
0.0841	87.2	0.4029	222.4	0.7669	133.7
0.1270	124.3	0.4520	220.4	0.7675	133.5
0.1709	153.7	0.4524	222.2	0.8234	103.7
0.1712	151.5	0.5019	216.7	0.8806	73.0
0.2156	176.4	0.5529	207.1	0.8811	72.9
0.2612	194.0	0.6048	194.9		
$x(\text{CH}_3)_2\text{CO} + (1-x)\text{H}_2\text{O}$					
0.0062	-58.0	0.0946	-547.1	0.5346	-72.4
0.0127	-116.9	0.1162	-596.6	0.5795	14.0
0.0127	-130.7	0.1399	-619.3	0.5802	22.9
0.0127	-119.4	0.1966	-634.8	0.6300	109.4
0.0264	-229.3	0.1962	-644.1	0.6863	200.7
0.0413	-326.3	0.3115	-514.0	0.7497	278.6
0.0413	-339.1	0.3123	-503.0	0.8205	312.4
0.0575	-415.4	0.4229	-299.5	0.8218	319.8
0.0753	-489.4	0.4229	-304.7	0.9036	264.9
0.0754	-502.9	0.4935	-157.8		
$x\text{CH}_3\text{OH} + (1-x)\text{H}_2\text{O}$					
0.0470	-301.9	0.3082	-883.9	0.7990	-461.3
0.1000	-560.1	0.4002	-853.6	0.8927	-284.8
0.1599	-739.2	0.5089	-782.8		
0.2285	-841.1	0.6395	-675.5		
$x\text{CHCl}_3 + (1-x)\text{c-(CH}_2)_5\text{O}$					
0.3416	-2268.7	0.4974	-2666.6	0.6919	-2298.1
$x\text{CHCl}_3 + (1-x)\text{c-(CH}_2)_5\text{NH}$					
0.0614	-662.7	0.4495	-3263.9	0.7407	-2216.7
0.1205	-1285.4	0.5005	-3263.4	0.7859	-1896.1
0.1783	-1838.5	0.5504	-3177.8	0.8304	-1543.9
0.2349	-2304.6	0.5505	-3172.0	0.8740	-1177.6
0.2902	-2709.0	0.5995	-3016.6	0.9168	-797.3
0.3444	-3001.1	0.6475	-2761.2	0.9588	-420.9
0.3975	-3180.6	0.6946	-2513.6		

TABLE 5—continued

x	$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J} \cdot \text{mol}^{-1}}$
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}$					
0.0467	-460.4	0.3308	-2822.4	0.6815	-2785.4
0.0931	-921.1	0.3797	-3046.3	0.7334	-2450.2
0.1399	-1367.0	0.4288	-3193.7	0.7857	-2048.0
0.1869	-1777.4	0.4784	-3294.8	0.8386	-1604.2
0.2345	-2173.7	0.5285	-3331.7	0.8919	-1077.4
0.2824	-2521.8	0.5793	-3249.2	0.9457	-589.2
$x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{NH}$					
0.0941	-1235.0	0.3826	-3735.1	0.6842	-2889.3
0.1413	-1817.4	0.4319	-3804.1	0.7359	-2494.4
0.1889	-2338.4	0.4816	-3809.4	0.7878	-2056.2
0.2368	-2834.8	0.5317	-3705.5	0.8403	-1590.4
0.2850	-3233.1	0.5821	-3499.6	0.8931	-1079.9
0.3336	-3517.7	0.6330	-3219.4	0.9463	-563.5

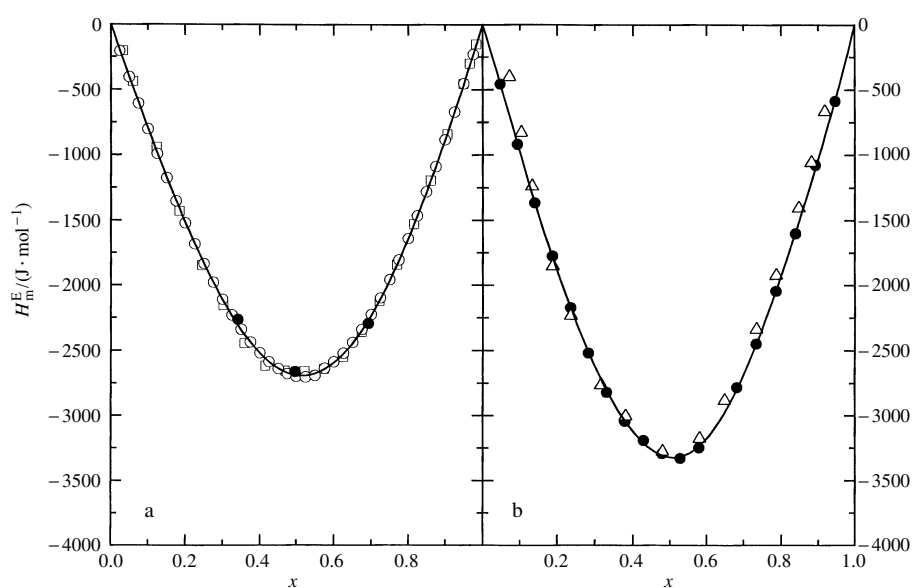


FIGURE 8. Excess molar enthalpies H_m^E at 298.15 K plotted against mole fraction x for a, $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{O}\}$; and b, $\{x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}\}$. ●, This work; ○, Becker and Kiefer;⁽³⁴⁾ □, Inglese;⁽³⁵⁾ △, Francesconi and Comelli;⁽³⁶⁾ —, fit by equation (5) of a, data of Becker and Kiefer⁽³⁴⁾ and b, data from this work.

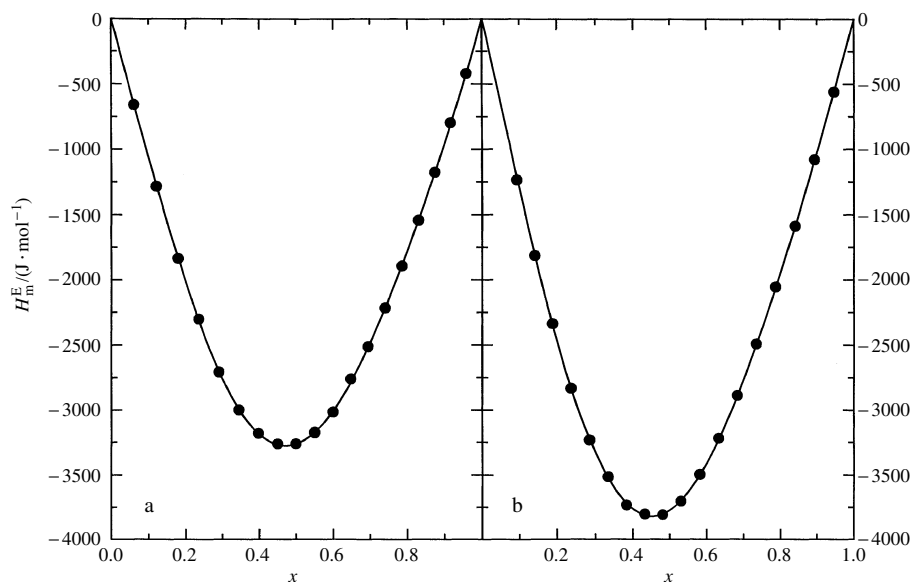


FIGURE 9. Excess molar enthalpies H_m^E at 298.15 K plotted against mole fraction x for a, $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{NH}\}$ and b, $\{x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{NH}\}$: ●, this work; —, fit by equation (5).

numerous and precise H_m^E data exist in the literature,^(34,35) we confined ourselves to confirmation experiments at just three compositions ($x \approx 0.3, 0.5$ and 0.7). As seen from figure 8(a), all these measurements are in excellent agreement. For $\{x\text{CF}_3\text{CHBrCl} + (1-x)c\text{-(CH}_2)_5\text{O}\}$, H_m^E data have been also reported previously.⁽³⁶⁾ Figure 8(b) shows that our H_m^E values agree well with the literature for $0.15 < x < 0.65$, but outside this range the literature H_m^E deviate slightly but systematically, being less negative. A closer inspection indicates that the data of Francesconi *et al.*⁽³⁶⁾ imply the existence of inflection points on $H_m^E(x)$ near both dilution limits. As such a situation is much more likely an artefact due to experimental error than true behaviour, we consider the present data to be superior to those of reference 36.

The measured values of H_m^E for each of the mixtures with the exception of $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_5\text{O}\}$ were fitted with the smoothing equation:

$$H_m^E/(\text{J} \cdot \text{mol}^{-1}) = x(1-x) \sum_{i=1}^q b_i (2x-1)^{(i-1)}. \quad (5)$$

The parameters b_i determined by the unweighted least-squares method together with the standard deviation of fit s are given in table 6. The number of parameters in equation (5) was chosen for each system on the basis of the same statistical criteria as in the case of equation (4).

The observed very large negative values of G_m^E and H_m^E indicate that, for all trichloromethane- or halothane-containing mixtures under study, the extent of complex

TABLE 6. Parameters b_i and standard deviations for representation of excess molar enthalpies at 298.15 K by equation (5)

	b_1	b_2	b_3	b_4	b_5	b_6	$s^a / (\text{J} \cdot \text{mol}^{-1})$
$x\text{CHCl}_3 +$ $(1-x)\text{c}-(\text{CH}_2)_5\text{NH}$	-13048.7	1976.4	3693.2	-1845.8	-1397.1		9.1
$x\text{CF}_3\text{CHBrCl} +$ $(1-x)\text{c}-(\text{CH}_2)_5\text{O}$	-13290.0	-641.6	4238.9	2416.8	-1423.1	-2809.9	14
$x\text{CF}_3\text{CHBrCl} +$ $(1-x)\text{c}-(\text{CH}_2)_5\text{NH}$	-15097.5	3778.8	3509.9	-2880.8			9.7
$x\text{C}_6\text{H}_{14} +$ $(1-x)\text{c}-\text{C}_6\text{H}_{12}$	867.1	-258.2	68.1	-33.4			0.9
$x(\text{CH}_3)_2\text{CO} +$ $(1-x)\text{H}_2\text{O}$	-580.2	4176.4	-641.8	1392.0	-1567.3	1840.0	5.4
$x\text{CH}_3\text{OH} + (1-x)\text{H}_2\text{O}$	-3173.7	1524.2	-2212.3	768.0			5.6

$$^a s = \left\{ \sum_{j=1}^n (\Delta H_{m,j}^E)^2 / (n-q) \right\}^{1/2}.$$

formation and the stability of complexes are relatively high. Comparisons of the excess quantities for different mixtures confirm our previous finding^(1,4) that halothane is a stronger proton donor than trichloromethane and, in addition, suggest that piperidine is a more efficient proton acceptor than tetrahydropyran. Inferences from the non-ideality of vapour-phase mixtures, as expressed by the Hayden–O’Connell solvation parameters η_{ij} lead to the same conclusions. Looking closely at G_m^E and H_m^E behaviour depicted in figures 3 and 4 and 8 and 9 one notes another distinct difference between the tetrahydropyran- and piperidine-containing mixtures: while the former are nearly exactly symmetric, the latter have their extrema shifted toward lower proton donor contents. Taking into account the structure of the piperidine molecule, this shift might be hypothesised as being an indication of appreciable piperidine self-association. A detailed theoretical analysis of the G_m^E and H_m^E data and of additional data on excess molar heat capacities and n.m.r. shifts will be presented in a forthcoming publication⁽¹²⁾ where we will characterise the complex formation in systems under study quantitatively and examine the effect of piperidine self-association through application of association models.

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